

Computer Networks Tutorial

Tutorial-1

1. Consider a channel with source and destination having distance between them of 10km. Propagation delay is 10 μ sec/km. What will be the data rate of the channel if it is given that the RTT is equal to the transmission delay and packet size is of 125 bytes?
2. The following data fragment occurs in the middle of a data stream for which the byte-stuffing algorithm described in the text is used: A B ESC C ESC FLAG FLAG D. What is the output after stuffing?
3. A bit string, 011110111110111110, needs to be transmitted at the data link layer. What is the string actually transmitted after bit stuffing?
4. When bit stuffing is used, is it possible for the loss, insertion, or modification of a single bit to cause an error not detected by the checksum? If not, why not? If so, how? Does the checksum length play a role here?

Tutorial-2

1. If the distance between A and B is 4000km, how long does it take computer A to receive ACK for a packet? Use the speed of light for propagation speed and assume time between receiving and sending ACK is zero.
2. A system uses the Stop-N-Wait protocol. If each packet carries 1000 bits of data, how long does it take to send 1 million bits of data if the distance between the sender and receiver is 5000 km and propagation speed is 2×10^8 m. Ignore transmission, waiting and processing delays. We assume no data or control frame is damaged or lost.
3. Two stations, A & B connected via point-to-point link and exchange frames using go-back-n sliding window protocol with window size 7. Sequence number is of 3 bits. Station A sends frame 0 to 6. B receives them in order but frame 4 was damaged. So what will be the buffer frames in the current window of A?
4. In stop and wait ARQ a station A sends 1000 bits packet to station B with 10,000 kbps. After sending a packet how much time computer will ideal if size of ACK is 2 bit and RTT us 1 sec.

Tutorial-3

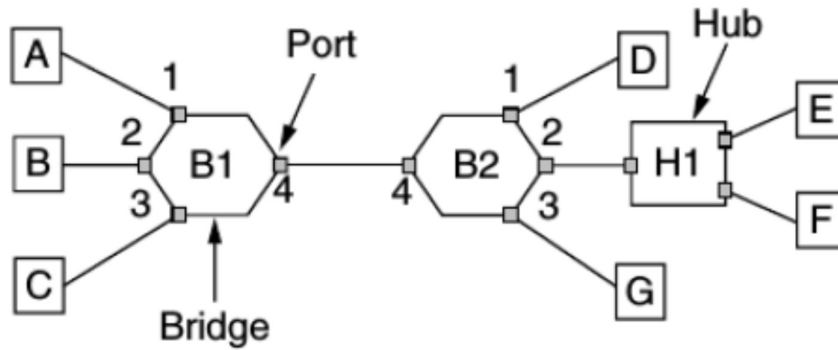
1. A 1000 km long cable operates at 1 Mbps. The propagation speed in the cable is $\frac{2}{3}$ the speed of the light. How many bits fit in the cable?
2. If the band width of the line is 1 Mbps, propagation delay is 20ms and packet size is 1kB then what will be the link utilization?
3. Consider the delay of pure ALOHA versus slotted ALOHA at low load. Which one is less?
4. A large population of ALOHA users manages to generate 50 requests/sec, including both originals and retransmissions. Time is slotted in units of 40 msec.
 - (a) What is the chance of success on the first attempt?
 - (b) What is the probability of exactly k collisions and then a success?

Tutorial-4

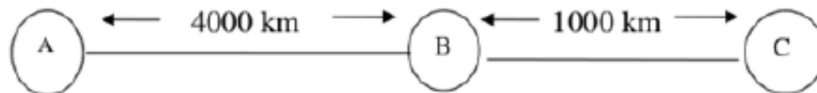
1. What is the length of a contention slot in CSMA/CD for (a) a 2-km twin-lead cable (signal propagation speed is 82% of the signal propagation speed in vacuum), and (b) a 40-km multimode fiber optic cable (signal propagation speed is 65% of the signal propagation speed in vacuum)?
2. How long does a station s , have to wait in the worst case before it can start transmitting its frame over a LAN that uses the basic bit-map protocol?
3. An IP packet to be transmitted by Ethernet is 60 bytes long, including all its headers. If LLC is not in use, is padding needed in the Ethernet frame, and if so, how many bytes?
4. Ethernet frames must be at least 64 bytes long to ensure that the transmitter is still going in the event of a collision at the far end of the cable. Fast Ethernet has the same 64-byte minimum frame size but can get the bits out ten times faster. How is it possible to maintain the same minimum frame size?
5. A 1-km-long, 10-Mbps CSMA/CD LAN (not 802.3) has a propagation speed of 200 m/ μ sec. Repeaters are not allowed in this system. Data frames are 256 bits long, including 32 bits of header, checksum, and other overhead. The first bit slot after a successful transmission is reserved for the receiver to capture the channel in order to send a 32-bit acknowledgement frame. What is the effective data rate, excluding overhead, assuming that there are no collisions?
6. Sixteen stations, numbered 1 through 16, are contending for the use of a shared channel by using the adaptive tree walk protocol. If all the stations whose addresses are prime numbers suddenly become ready at once, how many bit slots are needed to resolve the contention?
7. Consider five wireless stations, A,B,C, D, and E. Station A can communicate with all other stations. B can communicate with A, C and E. C can communicate with A, B and D. D can communicate with A, C and E. E can communicate A,D and B. (a) When A is sending to B, what other communications are possible? (b) When B is sending to A, what other communications are possible? (c) When B is sending to C, what other communications are possible?
8. Six stations, A through F, communicate using the MACA protocol. Is it possible for two transmissions to take place simultaneously?

Tutorial-5

1. Suppose that an 11-Mbps 802.11b LAN is transmitting 64-byte frames back-to-back over a radio channel with a bit error rate of 10^{-7} . How many frames per second will be damaged on average?
2. Bluetooth uses FHSS with 1600 hops per second. What is the dwell time?
3. Suppose the hash tables in the two bridges are empty. List all ports on which a packet will be forwarded for the following sequence of data transmissions:
 - a) A sends a packet to C.
 - b) E sends a packet to F.
 - c) F sends a packet to E.
 - d) G sends a packet to E.
 - e) D sends a packet to A.
 - f) B sends a packet to F.



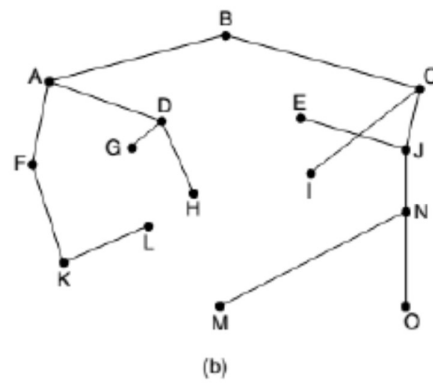
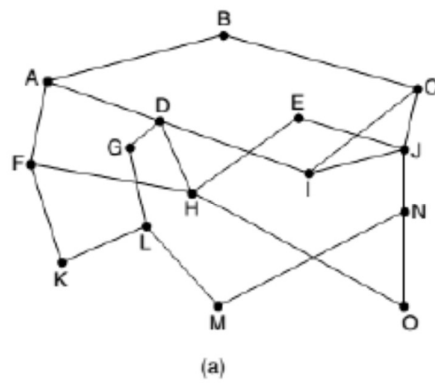
4. In following figure frames are generated at node A and sent to node C through node B. Determine the minimum transmission rate required between nodes B and C so that the buffers at node B are not flooded, based on the following:
- The data rate between A and B is 100 kbps.
 - The propagation delay is 5 $\mu\text{sec}/\text{km}$ for both lines
 - There are full-duplex, error-free lines between the nodes.
 - All data frames are 1000 bits long; ACK frames are separate frames of negligible length.
 - Between A and B, a sliding window protocol is used, with a window size of 3 (three).
 - Between B and C, stop and wait is used.



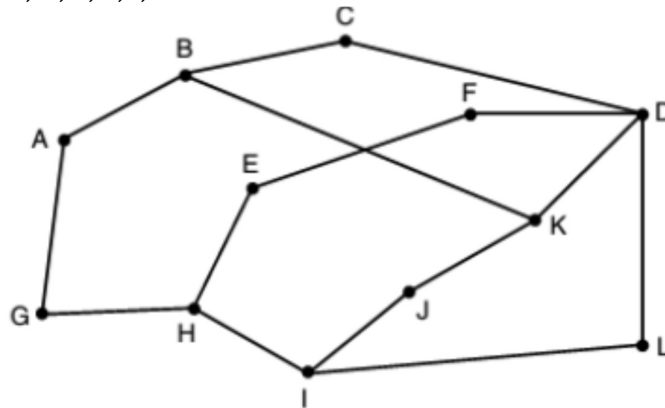
5. A 4Mbps token ring has a token holding time value of 10msec. What is the longest frame that can be sent on this ring?
6. Ethernet frames must be at least 64 bytes long to ensure that the transmitter can detect collisions. A faster Ethernet has the same minimum frame size but can transmit 10 times faster. How is it possible to still detect collisions?
7. Assume CSMA/CD protocol. Find the minimum frame length for a 1Mbps bit rate and maximum network span of 10 km with no repeaters. Assume a medium propagation delay of 4.5 nanoseconds per meter. Is CSMA/CD a reasonable protocol for a network of this span and bit rate.
8. Assuming that all routers and hosts are working properly and that all software in both is free of all errors, is there any chance, however small, that a packet will be delivered to the wrong destination?

Tutorial-6

1. How many packets are generated by a broadcast from B, using (a) reverse path forwarding? (b) the sink tree?



2. Compute a multicast spanning tree for router C for a group with members at routers A, B, C, D, E, F, I, K



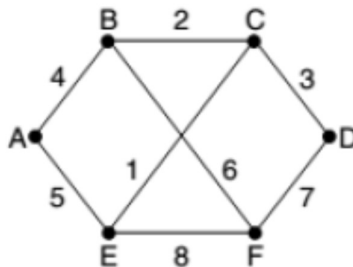
3. Suppose that host A is connected to a router R 1, R 1 is connected to another router, R 2, and R 2 is connected to host B. Suppose that a TCP message that contains 900 bytes of data and 20 bytes of TCP header is passed to the IP code at host A for delivery to B. Show the Total length, Identification, DF, MF, and Fragment offset fields of the IP header in each packet transmitted over the three links. Assume that link A-R1 can support a maximum frame size of 1024 bytes including a 14-byte frame header, link R1-R2 can support a maximum frame size of 512 bytes, including an 8-byte frame header, and link R2-B can support a maximum frame size of 512 bytes including a 12-byte frame header.
4. A 4480 octet datagram is to be transmitted and needs to be fragmented because it will pass through an Ethernet with a maximum payload of 1500 octets. Show the Total Length, More flag, Fragment Offset values in each of the resulting fragments.
5. Given the network address 133.22.0.0, find the class, the network id, and the range of the addresses.
6. What is the subnetwork address if the destination address is 200.45.34.56 and the subnet mask is 255.255.240.0?
7. A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets. Find the number of addresses of each subnet.
8. You currently use the default mask for your IP network 192.168.1.0. You need to subnet your network so that you have 30 additional networks, and 4 hosts per network. Is this possible, and what subnet mask should you use?

Tutorial-7

1. A small organization is given a block with the beginning address and the prefix length 205.16.37.24/29 (in slash notation). What is the range of the block?
2. A small organization is given a block with the beginning address and the prefix length 205.16.37.24/29 (in slash notation). What is the range of the block?
3. In the TCP/IP model, which layer is responsible for logical addressing of host computers and routing messages? Say as much as you can about how the following sequences identify a particular network, host computer and application, assuming the classfull addressing system:
 - 129.8.45.13:25
 - 124.42.5.45:23
 - 220.3.6.23:80
4. We need to make a supernet out of 16 class C blocks. What is the supernet mask?
5. A router is blasting out IP packets whose total length (data plus header) is 1024 bytes. Assuming that packets live for 10 sec, what is the maximum line speed the router can operate at without danger of cycling through the IP datagram ID number space?
6. Suppose that instead of using 16 bits for the network part of a class B address originally, 20 bits had been used. How many class B networks would there have been?
7. While IP addresses are tied to specific networks, Ethernet addresses are not. Can you think of a good reason why they are not?
8. A large number of consecutive IP addresses are available starting at 198.16.0.0. Suppose that four organizations, A, B, C, and D, request 4000, 2000, 4000, and 8000 addresses, respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the w.x.y.z/s notation.

Tutorial-8

1. Given the following IP in CIDR format 192.167.12.0/23. Find the network, the host addresses, and the range of addresses.
2. Distance vector routing is used, and the following vectors have just come in to router C: from B: (5, 0, 8, 12, 6, 2); from D: (16, 12, 6, 0, 9, 10); and from E: (7, 6, 3, 9, 0, 4). The cost of the links from C to B, D, and E, are 6, 3, and 5, respectively. What is C's new routing table? Give both the outgoing line to use and the cost.



3. If costs are recorded as 8-bit numbers in a 50-router network, and distance vectors are exchanged twice a second, how much bandwidth per (full-duplex) line is chewed

up by the distributed routing algorithm? Assume that each router has three lines to other routers.

4. For hierarchical routing with 4800 routers, what region and cluster sizes should be chosen to minimize the size of the routing table for a three-layer hierarchy? A good starting place is the hypothesis that a solution with k clusters of k regions of k routers is close to optimal, which means that k is about the cube root of 4800 (around 16). Use trial and error to check out combinations where all three parameters are in the general vicinity of 16.
5. A router has just received the following new IP addresses: 57.6.96.0/21, 57.6.104.0/21, 57.6.112.0/21, and 57.6.120.0/21. If all of them use the same outgoing line, can they be aggregated? If so, to what? If not, why not?
6. The set of IP addresses from 29.18.0.0 to 29.18.128.255 has been aggregated to 29.18.0.0/17. However, there is a gap of 1024 unassigned addresses from 29.18.60.0 to 29.18.63.255 that are now suddenly assigned to a host using a different outgoing line. Is it now necessary to split up the aggregate address into its constituent blocks, add the new block to the table, and then see if any re-aggregation is possible? If not, what can be done instead?
7. A router has the following (CIDR) entries in its routing table:
Address/mask Next hop
135.46.56.0/22 Interface 0
135.46.60.0/22 Interface 1
192.53.40.0/23 Router 1
default Router 2
For each of the following IP addresses, what does the router do if a packet with that address arrives? (a) 135.46.63.10 (b) 135.46.57.14 (c) 135.46.52.2 (d) 192.53.40.7 (e) 192.53.56.7
8. Consider an application that transmits data at a steady rate (for example the sender generates an N -bit unit of data every K time units, where K is small and fixed). Also, when such an application starts, it will continue running for a relatively long period of time. Answer the following questions, briefly justifying your answers. a) Would a packet-switched network or a circuit-switched network be more appropriate for this application? Why? b) Suppose a packet-switched network is used and the only traffic in this network comes from this application as described above. Furthermore, assume that the sum of the application data rates is less than the capacities of each and every link. Is some form of congestion control needed? Why?

Tutorial-9

1. Imagine a flow specification that has a maximum packet size of 1000 bytes, a token bucket rate of 10 million bytes/sec, a token bucket size of 1 million bytes, and a maximum transmission rate of 50 million bytes/sec. How long can a burst at maximum speed last?
2. In a leaky bucket used to control liquid flow, how many gallons of liquid are left in the bucket if the output rate is 5 gal/min, there is an input burst of 100 gal/min for 12 s, and there is no input for 48 s?
3. A computer on a 6-Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 1 Mbps. It is initially filled to capacity with 8 megabits. How long can the computer transmit at the full 6 Mbps?

4. A token bucket scheme is used for traffic shaping. A new token is put into the bucket every 5 μ sec. Each token is good for one short packet, which contains 48 bytes of data. What is the maximum sustainable data rate?

Tutorial-10

1. Consider the Parameters:

N = Number of Hops between two given system = 4

L = Message Length in bits = 3200

B = Data rate in BPS = 9600

P = Fixed Packet size in Bits = 1024

H = Overhead Bits per Packet = 16

S = Call setup time in Second = 0.2

D = Propagation Delay per hop in second. = 0.001

Find End to End Delay For circuit switching only.

2. A client has a packet of 68,000 bytes. Show how this packet can be transferred by using only one UDP user datagram
3. A client uses UDP to send data to a server. The data are 16 bytes. Calculate the efficiency of this transmission at the UDP level (ratio of useful bytes to total bytes).
4. The following is a dump of a UDP header in hexadecimal format. 0632000D001CE217
- What is the source port number?
 - What is the destination port number?
 - What is the total length of the user datagram?
 - What is the length of the data?
 - Is the packet directed from a client to a server or vice versa?
 - What is the client process?
5. In a leaky bucket used to control liquid flow, how many gallons of liquid are left in the bucket if the output rate is 5 gal/min, there is an input burst of 100 gal/min for 12s, and there is no input for 48 s?
6. TCP is sending data at 1 M Byte/s. If the sequence number starts with 7000, how long does it take before the sequence number goes back to zero?